

Bayesian Estimator of Selfing (BES)

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1 Introduction

BES is a software package for estimating self-fertilization (selfing) rates and other mating system parameters from genotype data. BES estimates parameters in a Bayesian framework using Markov chain Monte Carlo (MCMC). BES contains models of pure hermaphroditism, androdioecy (hermaphrodites + males), and gynodioecy (hermaphrodites + females). Under each model, BES estimates selfing rates, mutation rates, and mating-system specific parameters. BES also contains a generic model for estimating selfing rates and mutation rates independent of a mating system. Additional non-genetic information, such as field observations of the number of females or males, is required for estimating parameters under the gynodioecious model and the androdioecious model.

BES runs through BALi-Phy in a terminal on Linux, macOS, and Windows. It is an extension package for BALi-Phy and requires version 4.3 or later. See also the BALi-Phy user guide.

BES contains a number of modules that correspond to different mating system models. Each model allows estimating a different set of parameters. The generic model and the pure hermaphrodite model without inbreeding depression can be run without modification to estimate the selfing rate and locus-specific mutation rates.

However, the gynodioecious and androdioecious models require additional information besides the genetic data. The corresponding scripts require observed male or female counts on the command line. Some models also require the user to edit the module to choose priors, fix parameters, or add other identifying information. This manual describes how to add information, but is not a substitute for understanding the structure of the model.

2 Installation

2.1 Installing BALi-Phy

Since BES is an extension package for BALi-Phy, you must first install BALi-Phy before you can use BES. To install BALi-Phy, follow the installation instructions for BALi-Phy.

2.2 Installing additional software

For graphical inspection of TSV parameter logs, optionally install:

- Tracer helps to visualize the results of MCMC runs.

2.3 Installing BES

First, check that the `bali-phy-pkg` command works:

```
% bali-phy-pkg help
```

Download and install the BES package:

```
% bali-phy-pkg install BES
```

Check that the package is installed:

```
% bali-phy-pkg packages
```

To see what modules were installed, run:

```
% bali-phy-pkg files BES
```

You can uninstall the package by running:

```
% bali-phy-pkg uninstall BES
```

The package supplies reusable modules. Clone the repository to obtain the editable model templates and example data, then run the examples from that directory:

```
git clone https://github.com/bredelings/BayesianEstimatorSelfing.git
cd BayesianEstimatorSelfing
```

All supplied models estimate the additional non-selfing inbreeding component described below. The ready-to-run models are `Generic.hs`, `Generic2.hs`, `Andro.hs`, `Examples/Andro.hs`, `Examples/Gyno.hs`, and the installed `PopGen.Selfing.Herm` model. The androdioecious and gynodioecious models require observed sex counts as command-line arguments.

`HermID.hs`, `AndroID.hs`, and top-level `Gyno.hs` deliberately leave parameter definitions commented out. They must be edited before execution: supply priors, fixed values, or observations appropriate to the analysis. `Examples/Gyno.hs` supplies an illustrative prior on relative selfed-seed viability and fixes relative female seed production to 1. Review these assumptions before using it for your data. The two runnable androdioecy scripts also contain different prior choices.

3 Running the program

3.1 Quick Start

First, inspect the initial model values without starting MCMC or creating an output directory:

```
bali-phy run Generic.hs -l tsv --test Examples/outfile.001.70.001.phase1
```

Then run a short demonstration analysis using the generic model:

```
bali-phy run Generic.hs -l tsv --iterations=1000 Examples/outfile.001.70.001.phase1
```

These 1,000 iterations demonstrate execution; they do not establish convergence. Assess mixing and convergence before interpreting results. The run creates a directory called `Generic-1/` (or `Generic-2/`, etc.) that contains the output files.

After the run finishes, load the output file `Generic-1/C1.log` using the GUI program Tracer. On Unix, you can run this from the command line as follows:

```
% tracer Generic-1/C1.log &
```

It is also possible to use a non-graphical program `statreport` to view the estimates of the selfing rate

```
% statreport --select="s*" Generic-1/C1.log
```

This can be useful when analyzing data in a terminal.

4 Input

`Generic.hs` and the mating-system models read the PHASE layout described below. Specify both alleles for each locus, using `NA` for missing data.

`Generic2.hs` instead reads FastPhase or Phase2 files. The supplied examples can be inspected with:

```
bali-phy run Generic2.hs -l tsv --test Examples/test.fastphase
bali-phy run Generic2.hs -l tsv --test Examples/test.phase2
```

Use `--iterations` instead of `--test` to sample from these models. The files in `Examples/` illustrate the accepted layouts.

A PHASE file contains a 3-line header, followed by a single line for each observed individual. The header consists of

1. The number of individuals, on a line by itself.
2. The number of loci, on a line by itself.
3. A sequence of 'M's (for microsatellite) on a line by itself. The number of M's should equal the number of loci.

The line describing each individual should contain an individual name, followed by a list of integer allele names. The name and the numbers should be tab-delimited, and there should be twice the number of alleles as loci, since there are 2 alleles per locus. Integer allele names must be positive. The purpose of this is to avoid confusion, since 0 and negative numbers are often used to indicate missing data.

Here is a very small PHASE file as an illustration:

```
2
3
MMM
sample.1    23  23  2   1   NA  NA
sample.2    23  20  1   1   4   5
```

5 Output

5.1 Output directory

For an MCMC run, BALI-Phy creates a new directory to store its output files. By default, its base name is the name of the model, with a number added to make the directory unique. For example, the **Generic** model uses **Generic-1/**, then **Generic-2/**, and so on.

The **--name** option selects a different base name while retaining automatic unique-directory creation. Alternatively, **--output-dir** writes into the exact directory specified; that directory must already exist. If both options are supplied, **--output-dir** takes precedence. A run with **--test** does not create an output directory.

5.2 Output files

The selected logging format determines the parameter files:

File name	Description
C1.log.json	JSON parameter samples, written by default.
C1.log	Tab-separated parameter samples, written with -l tsv .

Use **-l json,tsv** to write both formats. The scripts print progress and diagnostics to the terminal; they do not automatically create **C1.out** or **C1.err**. A **--test** invocation prints initial values rather than creating parameter logs.

The logs contain samples of model parameters and derived quantities. For example, **s*** is the fraction of uniparental individuals at the time of breeding. Use TSV output for Tracer and the **statreport** commands below.

Since each mating system has a unique set of parameters, the variables for each model will be described in the section for that model.

6 Analyzing output files

6.1 Using `statreport`

In order to determine estimates of parameters, you can use the program `statreport`:

```
% statreport C1.log
```

To select particular parameters for numerical summaries, use:

```
% statreport --select "s*" C1.log
```

Here quotes are necessary to make sure that the `*` is not interpreted by the command line shell, but is passed in to the `statreport` program unchanged.

You can also use the arguments `--mean`, `--mode`, and `--median` (the default) to examine different properties of the posterior distribution.

6.2 Using `tracer`

Tracer is a graphical program for exploring posterior distributions.

Run the model with `-l tsv`, then load the resulting `C1.log` file using `tracer`. On Unix, if `tracer` is in your `PATH`, you can do this by typing

```
% tracer C1.log &
```

7 Mating system models

The stochastic process that generates the genetic data is described in terms of mating system parameters Ψ and the effective scaled mutation rates Θ_l^* for each locus l . The set of variables Ψ is specific for each mating system.

Some quantities are of interest for each mating system

- s^* is the fraction of uniparental adults (selfing rate).
- R is the decrease in effective population size caused by the mating system.
- T_k is the number of generations of selfing in the immediate ancestry of individual k .
- Θ_l is the scaled mutation rate $4Nu$ for locus l .
- Θ_l^* is the *effective* scaled mutation rate for locus l .

Here, R is given by

- $R = \lim_{N \rightarrow \infty} \frac{N^*}{N}$

where N is the population size, and N^* is the effective population size defined by the rate of parent-sharing. The effective scaled mutation rate is

- $\Theta_l^* = \Theta_l \cdot (1 - s^*/2) \cdot R$.

In general, s^* and R are composite parameters: they are determined from the basic mating system parameters Ψ .

It is important to note that the genetic data contain information about s^* , Θ_l^* , and the additional inbreeding component described below. Therefore, the basic mating system parameters Ψ may be unidentifiable on the basis of genetic data alone. This can be solved by introducing additional data in the form of field observations on, for example, the fraction of males or the fraction of females in the population. When such information is unavailable, the generic model should be used.

7.1 Non-selfing inbreeding

All supplied model scripts use the robust estimator, which estimates $F[\text{other}]$ alongside selfing. This allows decreased heterozygosity to have sources other than selfing. The model uses a shared $F[\text{other}]$ across individuals, while individual selfing histories vary. Information across loci helps distinguish these components, subject to the data and model assumptions.

For an individual without recent selfing, its two alleles are independent draws from the gene pool with probability $1 - F_{\text{other}}$, or identical by descent with probability F_{other} . The latter case assumes rapid coalescence compared with an ordinary coalescent event, without specifying its mechanism. The supplied prior on $F[\text{other}]$ is `beta 0.25 1.0`.

The following fields supplement each model's parameter table below:

Field	Meaning
$F[\text{other}]$	Additional loss of heterozygosity, estimated from the model.
$F[\text{selfing}]$	Selfing component: $s^*/(2 - s^*)$.
$F[\text{total}]$	Combined component: $1 - (1 - F_{\text{selfing}})(1 - F_{\text{other}})$.

The mating-system parameter sets Ψ below describe reproduction. $F[\text{other}]$ is an additional parameter of the genetic observation model; the two other inbreeding coefficients are derived.

7.2 Generic model

The generic model is agnostic about the particular mating system, and reports s^* , T_k , Θ_l^* , and the common inbreeding fields as estimated from the genetic data. Since the mating system is not known, R cannot be calculated. This means that it is not possible to calculate Θ_l from Θ_l^* .

The generic model characterizes reproduction in terms of the selfing rate s^* . Thus, $\Psi = \{s^*\}$, and so s^* is a basic parameter, instead of being calculated from other parameters.

The following variables are estimated, with the field names given:

Variable	Name	Description
s^*	<code>s*</code>	Fraction of uniparental adults (selfing rate).
T_k	<code>t[k]</code>	Number of generations of selfing for individual k .
Θ_l^*	<code>theta*[l]</code>	<i>Effective</i> scaled mutation rate for locus l .

This variant is implemented by `Generic.hs` and `Generic2.hs`, which differ in their input format.

7.3 Pure Hermaphrodite

In the pure hermaphrodite model, each individual contributes both male and female gametes to the gene pool. There are two variants of this model.

7.3.1 Variant I

The first variant has $\Psi = \{s^*\}$. This variant treats s^* as a basic parameter, and does not model inbreeding depression. This variant is similar to the generic model, except that R can be calculated because the mating system is known to be a pure hermaphrodite mating system. This allows Θ_l to be calculated also. However, note that R is always 1 for the pure hermaphrodite model.

The following variables are estimated, with the field names given:

Variable	Name	Description
s^*	s^*	Fraction of uniparental adults (selfing rate).
T_k	$t[k]$	Number of generations of selfing for individual k .
R	R	Decrease in parent-sharing effective population size
Θ_l^*	$\text{theta}^*[l]$	<i>Effective</i> scaled mutation rate for locus l .
Θ_l	$\text{theta}[l]$	Scaled mutation rate $4Nu$ for locus l .

This variant is implemented by the installed `PopGen.Selfing.Herm` model.

7.3.2 Variant II

The second variant has $\Psi = \{\tilde{s}, \tau\}$. This variant treats s^* as a composite parameter.

The following variables are estimated, with the field names given:

Variable	Name	Description
$\tilde{s}$	$s\sim$	Fraction of uniparental seeds.
τ	tau	Relative viability of selfed seeds.
s^*	s^*	Fraction of uniparental adults (selfing rate).
T_k	$t[k]$	Number of generations of selfing for individual k .
R	R	Decrease in parent-sharing effective population size
Θ_l^*	$\text{theta}^*[l]$	<i>Effective</i> scaled mutation rate for locus l .
Θ_l	$\text{theta}[l]$	Scaled mutation rate $4Nu$ for locus l .

Here the user must modify `HermID.hs` to add additional information about $\tilde{s}$ or τ .

This variant is implemented by `HermID.hs`.

7.4 Androdioecy

In the androdioecious model, the population consists of some fraction p_m of males, with the rest of the individuals being hermaphrodites. Hermaphrodites produce male gametes, but only fertilize their own eggs. There are two variants of this model.

7.4.1 Variant I

The first variant has $\Psi = \{s^*, p_m\}$. This variant treats s^* as a basic parameter, and does not model inbreeding depression.

The following variables are estimated, with the field names given:

Variable	Name	Description
s^*	s^*	Fraction of uniparental adults (selfing rate).
p_m	p_m	The fraction of males
T_k	$t[k]$	Number of generations of selfing for individual k .
R	R	Decrease in parent-sharing effective population size
Θ_l^*	$\text{theta}^*[l]$	<i>Effective</i> scaled mutation rate for locus l .
Θ_l	$\text{theta}[l]$	Scaled mutation rate $4Nu$ for locus l .

The observed number of males supplies additional information about p_m through a binomial likelihood. Supply the number of males and the total number of surveyed individuals as required model arguments:

```
% bali-phy run Andro.hs --males 20 --total 2000 data.phase
```

These commands assume that the named input file exists. The priors remain specified in `Andro.hs` and must be edited there if different priors are desired.

7.4.2 Variant II

The second variant has $\Psi = \{\tilde{s}, \tau, p_m\}$. This variant treats s^* as a composite parameter.

The following variables are estimated, with the field names given:

Variable	Name	Description
$\tilde{s}$	s~	Fraction of uniparental zygotes.
τ	tau	Relative viability of selfed zygotes.
p_m	p_m	The fraction of males
s^*	s*	Fraction of uniparental adults (selfing rate).
T_k	t[k]	Number of generations of selfing for individual k .
R	R	Decrease in parent-sharing effective population size
Θ_l^*	theta*[l]	<i>Effective</i> scaled mutation rate for locus l .
Θ_l	theta[l]	Scaled mutation rate $4Nu$ for locus l .

The required `--males` and `--total` arguments supply additional information about p_m . The user must still modify `AndroID.hs` to provide a prior or fixed value for $\tilde{s}$ or τ .

```
% bali-phy run AndroID.hs --males 20 --total 2000 data.phase
```

7.5 Gynodioecy

In the gynodioecious model, the population consists of some fraction p_f of females, with the rest of the individuals being hermaphrodites. Hermaphrodites produce pollen, and so can contribute male gametes to females, other hermaphrodites, and themselves.

In this model $\Psi = \{\tilde{s}, \tau, p_f, \sigma\}$. Here s^* is a composite parameter.

The following variables are estimated, with the field names given:

Variable	Name	Description
$\tilde{s}$	s~	Fraction of hermaphrodite seeds set by self-pollen.
τ	tau	Relative viability of selfed seeds.
p_f	p_f	The fraction of females
σ	sigma	Seed production rate of females relative to hermaphrodites.
s^*	s*	Fraction of uniparental adults (selfing rate).
T_k	t[k]	Number of generations of selfing for individual k .
R	R	Decrease in parent-sharing effective population size
H	H	Fraction of non-selfed individuals with a hermaphrodite seed-parent.
Θ_l^*	theta*[l]	<i>Effective</i> scaled mutation rate for locus l .
Θ_l	theta[l]	Scaled mutation rate $4Nu$ for locus l .

For a ready-to-run illustration, use `Examples/Gyno.hs`; it specifies the additional choices in its source. Top-level `Gyno.hs` is an incomplete template. The required `--females` and `--total` arguments supply additional information about p_f . The user must still modify `Gyno.hs` to provide priors or fixed values for two of the remaining three components of Ψ .

```
% bali-phy run Gyno.hs --females 27 --total 221 data.phase
```

8 Specifying additional information

When the mating system parameters Ψ contain more than one degree of freedom, the mating system parameters are not identifiable from genetic data alone. The androdioecious and gynodioecious scripts incorporate observed sex counts supplied on the command line. Other observations, fixed values, and prior choices are specified by modifying the model-description module (`HermID.hs`, `Andro.hs`, `AndroID.hs`, or `Gyno.hs`). In general, if Ψ contains n variables, then additional information about $n - 1$ of them must be incorporated.

8.1 Modifying model-description modules

BES model-description modules (such as `Andro.hs`) use a Haskell-like syntax. In this syntax,

1. Function application $f(x,y)$ is written `f x y`.
2. Comments are introduced by `--`.
3. In a `do` block, an `observe` command introduces data. Commenting out the `observe` statements removes the data from the model file.

In the model-description template files (e.g. `HermID.hs`), the comments illustrate possible ways to introduce variables.

8.2 Methods for adding additional information

Additional information about a variable can be added in 3 ways.

1. Add observations that depend on that variable.
2. Fix the variable to a known constant value.
3. Place a subjective prior on the variable.

The supplied androdioecious and gynodioecious scripts already implement the common case of a binomial observation on the number of males or females. These counts are required command-line arguments and should not also be added to the source. The following source modifications apply to other observations and parameters.

8.2.1 Introduce a variable with a prior and place observations on it.

```
tau <- sample $ uniform 0.0 1.0
observe 10 $ binomial 20 $ toProb tau
```

8.2.2 Fix a variable to a known constant value.

If you know the value of a variable, you can fix it to a constant:

```
let tau = 1.0
```

If you have observations about a variable (e.g. p_m) then do not fix that variable to a constant. If you fix a variable to a constant, then that variable cannot be estimated since its value is already known.

8.2.3 Place a subjective prior on a variable

This approach doesn't actually make the parameter *identifiable*, since this approach affects only the prior, and not the likelihood.

```
tau <- sample $ beta 2.0 8.0
```

As a result, it is not possible to compare the posterior (with data) and the prior (without data) to assess the impact of the data. This approach is therefore not recommended.

9 Bayesian priors and posteriors

In Bayesian parlance, *prior* means “before the data”, and *posterior* means “after the data”. The Bayesian approach places prior distributions on parameters. As a result, parameters become random variables, and can have distributions. This differs from the maximum likelihood setting, where parameters are not random. However, the researcher must interpret prior and posterior distributions carefully in order to avoid drawing erroneous conclusions.

9.1 Uninformative priors

We take the “objective Bayesian” approach, and seek priors that have minimal influence on the analysis. In general, such priors have a broad range. This range should not be characterized simply in terms of the mean or median, but in terms of the Bayesian Credible Interval (BCI), and in terms of the shapes of the tails of the distribution.

Note that, although the goal is to obtain priors with minimal influence, such priors are not actually “uninformative”. Specifically, a uniform prior is not “uninformative”. The choice of an appropriate prior should be undertaken with care in order to avoid ruling out any plausible outcome *a priori*.

9.2 Comparing the posterior and the prior

The posterior distribution combines information from the prior distribution and the likelihood. In order to determine if the shape of the posterior is being largely driven by the prior, or largely driven by the likelihood, one should compare the posterior and the prior distributions.

9.3 Priors on composite parameters

For models, such as the gynodioecious model, where s^* is a composite parameter, no prior distribution is placed on s^* directly. However, s^* still has a prior distribution that is the combined result of the priors on all the basic parameters that it is computed from.

In order to determine what the shape of the prior on s^* is, one can run the model without data. This can be done by commenting out the `observe` statements.